

CHIARA-RENATA HORAK

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📍 Vienna, Austria

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EDUCATION

10/2024 - present **University of Vienna**

Master in Physics (Thesis: Symplectic Duality and Higgs Mechanisms in Supersymmetric Quivers)

10/2022 - 02/2026 **University of Vienna**

Bachelor in Mathematics (Thesis: Mathematical aspects of the Higgs Mechanism in the Standard Model of particle physics)

10/2021 - 09/2024 **University of Vienna**

Bachelor in Physics (Thesis: One-loop effects in Quantum Electrodynamics)

2013 - 2021

BRG 14 Linzerstraße, Vienna

Secondary School

EXTRACURRICULAR ACADEMIC ACTIVITIES

Vienna Master Class Mathematical Physics

Program for students who seek a rigorous education in mathematical physics

VDSP - Summer Academy 2025 - Symmetries, Particles and Fields

Vienna, 07/2025

Jürgen Ehlers Spring School on Gravitational Physics

Potsdam, 02-03/2024

Quantum Universe Attract.Workshop

Hamburg, 11/2025

TECHNICAL SKILLS

Languages

German (native)

English (fluent)

Latin (intermediate)

Programming

Python

Mathematica

Sage

Magma

Software

Microsoft 10 (Word, Excel, Outlook, Power Point), iOS

LaTeX

SciDAVis

Sophomore teaching

tutoring first year physics students

University tutor

tutoring first and second year physics students in Calculus I and Calculus III

Tutor

teaching university and high school students

ACHIEVEMENTS

- Merit scholarship for students with a GPA under 1.70 during a one year period

(1.49, academic year 2021/22)

(1.22, academic year 2022/23)

(1.35, academic year 2023/24)

(1.0, academic year 2024/25)

- Recieved language certificate C1 English at age 14

- Graduated high school at age 16 (with GPA 1.0)

TALKS

Symplectic duality and quiver algorithms

Quantum Universe Attract.Workshop, DESY 11/2025

Higgs Mechanisms in supersymmetric quiver gauge theories

Seminar in particle physics, University of Vienna 01/2026

Symplectic Duality and Higgs Mechanisms in Supersymmetric Quivers

Vienna Theory Lunch Seminar, University of Vienna 04/2026